1. (a) $f(x) = \frac{1}{2} x^T A x + b^T x$.

$x^T A x = \sum\sum x_i x_j A_{ij}$

$\frac{\partial}{\partial x_k}(x^T A x) = \frac{\partial}{\partial x_k}(\sum\sum x_i x_j A_{ij}) = 2\sum_{i=1}^{n} x_i A_{ki}$

$\frac{\partial}{\partial x_k}(b^T x) = b_k$

$\therefore \nabla f(x) = \begin{bmatrix} \sum x_i A_{i1} + b_1 \\ \sum x_i A_{i2} + b_2 \\ \vdots \\ \sum x_i A_{in} + b_n \end{bmatrix} = \begin{bmatrix} \sum x_i A_{i} \\ \vdots \\ \sum x_i A_{ni} \end{bmatrix} + b = Ax + b.$

(b) $\nabla f(x) = \nabla g(h(x)) = g'(h(x)) \cdot \begin{bmatrix} \frac{\partial}{\partial_1} h(x) \\ \vdots \\ \frac{\partial}{\partial_n} h(x) \end{bmatrix} = g'(h(x)) \nabla h(x)$

(c) $\nabla f(x) = Ax + b, \quad \nabla^2 f(x) = A$

(d) $\nabla g(a^T x) = g'(a^T x) \cdot \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = g'(a^T x) \cdot a, \quad \nabla^2 g(a^T x) = g''(a^T x) a a^T$

2 (a). $A = zz^T$

$A^T = (zz^T)^T = zz^T$, $\therefore$ Sym $(A = A^T)$

$x^TAx = x^Tzz^Tx$

$= (z^Tx)^T(z^Tx)$ z^Tx : 스칼라

$= (z^Tx)^2 \geq 0$

$\therefore A \succeq 0$ (PSD 임)

~~$N(A) = \{x \in \mathbb{R}^n : x \in Ax\}$~~

(b) $N(A) = \{x \in \mathbb{R}^n : Ax = 0\}$

$Ax = zz^Tx = (z^Tx)z$, $z \neq$ non-zero vector

(z^Tx : 스칼라)

$\therefore N(A) = \{x \in \mathbb{R}^n : z^Tx = 0\}$

$N(A) = z^\perp$

rank (A) = dim(span of A's col vector)

$= \dim(\text{span}\{z_1z, z_2z, \cdots, z_nz\})$

$= \dim(\text{span}\{z\})$

$= 1$.

(c) A is PSD.

$(BAB^T)^T = B(BA)^T = BA^TB^T = BAB^T$ $\therefore$ Sym

$x^T(BAB^T)x = (B^Tx)^TA(B^Tx)$ (B^Tx) is arbitrary n-vector

$= y^TAy \geq 0$.

BAB^T is PSD

3. (a)

$T = [t^{(1)} \cdots t^{(n)}]$

$T e_i = t^{(i)}$

$T^{-1} t^{(i)} = e_i$

$A t^{(i)} = (T \Lambda T^{-1}) t^{(i)} = T \Lambda (T^{-1} t^{(i)})$

$= T \Lambda e_i = T \begin{bmatrix} 0 \\ \lambda_i \\ 0 \end{bmatrix} = \lambda_i t^{(i)}$

(b) $A u^{(i)} = U \Lambda U^T u^{(i)} = U \Lambda e_i$

$= U \begin{bmatrix} 0 \\ \lambda_i \\ 0 \end{bmatrix} = \lambda_i u^{(i)}$

~~(c) A is PSD. $x^T A x$ on $u^{(i)}$ cases~~

~~$x^T A x = x^T U \Lambda U^T x$~~

(c) $x^T A x \geq 0$ for $\forall x \in \mathbb{R}^n$

$\therefore u^{(i)T} A u^{(i)} \geq 0.$

$\Leftrightarrow u^{(i)T} \lambda_i u^{(i)} \geq 0.$

$\Leftrightarrow \lambda_i (u^{(i)T} u^{(i)}) \geq 0 \quad u^T u = 1.$ (orthogonal)

$\Leftrightarrow \lambda_i \geq 0.$